Modeling with partial differential equations
Part III: the heat equation

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Definition of solutions to the heat equation

In this part, we will be interested in solutions to the following problem: Find $u : \mathbb{R}_+ \times \mathbb{R}^d \to \mathbb{R}$ (u depends on time and space $(t, x) \in \mathbb{R}_+ \times \mathbb{R}^d$) such that

$$\begin{cases} \partial_t u - \Delta u = f \\ \qquad u(0, \cdot) = u_0 \end{cases} \tag{1}$$

where, we recall $\Delta u = \sum_{i=1}^d \partial_{x_i, x_i} u$

We consider classical solutions: $f : \mathbb{R}_+ \times \mathbb{R}^d \to \mathbb{R}$ and $u_0 : \mathbb{R}^d \to \mathbb{R}$ are continuous, and u is in $\mathcal{C}(\mathbb{R}_+ \times \mathbb{R}^d) \cap \mathcal{C}^{1,2}(\mathbb{R}_+^* \times \mathbb{R}^d)$ i.e. continuous and $\mathcal{C}^1$ w.r.t to $t > 0$ and $\mathcal{C}^2$ w.r.t to $x \in \mathbb{R}^d$

Remark: The problem is linear with respect to (u_0, f). If u is solution to (1) for (u_0, f) and v is solution to (1) for (v_0, g), then a solution to (1) for $(u_0 + \lambda v_0, f + \lambda g)$ is $u + \lambda v$

Outline

- Derivation of the heat equation, and some applications
- Existence via the heat kernel
- Qualitative properties of the solutions to the heat equation
- Numerical approximation using finite differences

Two ingredients

We will need (see the previous lecture): gradient, divergence and the fundamental relation: For $u : \mathbb{R}^d \to \mathbb{R}$ in $\mathcal{C}^2$,

$$\operatorname{div}(\nabla u) = \sum_{i=1}^{d} \partial_{x_i}(\partial_{x_i} u) = \Delta u$$

We will also need the divergence formula: For $\Omega \subset \mathbb{R}^d$ (with regularity $\mathcal{C}^1$), and $F : \Omega \to \mathbb{R}^d$ a $\mathcal{C}^1(\overline{\Omega}, \mathbb{R}^d)$ vector field,

$$\int_\Omega \operatorname{div}(F) = \int_{\partial\Omega} F \cdot n \, d\sigma$$

where n is the unit outward normal to Ω and σ is the surface measure on $\partial\Omega$

Derivation of the heat equation

Let us consider a scalar quantity $u(x)$ (temperature, concentration of a chemical species) and the associated flux $F(x)$ (heat flux, species flux). Let us moreover assume that f is a bulk "source and sink" term

Step 1: Conservation law: over a time interval $(t, t + \Delta t)$, the evolution of the quantity u within a domain $\Omega \subset \mathbb{R}^d$ satisfies:

$$\int_\Omega u(t + \Delta t, x)\, dx - \int_\Omega u(t, x)\, dx$$
$$= \int_t^{t+\Delta t} \left(\int_\Omega f(s, x)\, dx - \int_{\partial\Omega} F(s, x) \cdot n(x) \sigma(dx) \right) ds$$

Remember the sign convention on F: the outgoing flux $F \cdot n$ is positive for a loss

Derivation of the heat equation

Divide by Δt and (formally) send $\Delta t \to 0$ to get the integral form of the conservation law: for any $\Omega \subset \mathbb{R}^d$,

$$\int_\Omega \frac{\partial u}{\partial t}(t,x)\,dx = \int_\Omega f(t,x)\,dx - \int_{\partial\Omega} F(t,x)\cdot n(x)\sigma(dx)$$

which yields, by the divergence theorem:

$$\int_\Omega \left(\frac{\partial u}{\partial t} - f + \operatorname{div}(F)\right)(t,x)\,dx = 0$$

Since Ω is arbitrary, one gets the differential form of the conservation law:

$$\frac{\partial u}{\partial t}(t,x) + \operatorname{div}(F)(t,x) = f(t,x)$$

Derivation of the heat equation

Step 2: Constitutive equation (Fourier law, Fick law): there exists $a > 0$ such that

$$F = -a\nabla u$$

Fluxes create flows from regions with large u to regions with small u, in order to "locally equilibrate" u

Conclusion: one thus obtains (since $\mathrm{div}\,(\nabla u) = \Delta u$): for $t > 0$ and $x \in \mathbb{R}^d$

$$\frac{\partial u}{\partial t}(t,x) - a\Delta u(t,x) = f(t,x)$$

As for an ordinary differential equation, this is supplemented with an initial condition at time $t = 0$: for $x \in \mathbb{R}^d$

$$u(0,x) = u_0(x)$$

Remark: Stationary solutions to the heat equation satisfy the Poisson equation $-a\Delta u = f$

The heat equation on a domain

As for the Laplace and Poisson equations one may be interested in the heat equation posed on an open bounded domain $\Omega \subset \mathbb{R}^d$, with boundary conditions:

Dirichlet problem: fix the value on the boundary

$$\begin{cases} \partial_t u - a\Delta u = f \text{ for } t > 0 \text{ and } x \in \Omega \\ u = g \text{ for } t > 0 \text{ and } x \in \partial\Omega \\ u(0,\cdot) = u_0 \text{ for } x \in \Omega \end{cases}$$

Neumann problem: fix the flux on the boundary

$$\begin{cases} \partial_t u - a\Delta u = f \text{ for } t > 0 \text{ and } x \in \Omega \\ \partial_n u = g \text{ for } t > 0 \text{ and } x \in \partial\Omega \\ u(0,\cdot) = u_0 \text{ for } x \in \Omega \end{cases}$$

Applications

The heat equation appears in numerous situations:

- Thermal diffusion: $\partial_t T - k\Delta T = f$
  (T is the temperature and k is the thermal diffusivity)
- Diffusion of chemical species: $\partial_t u - D\Delta u = f$
  (u is the concentration and D the diffusivity)

Remark: In machine learning, diffusion models generate new samples using a "time-reversed" heat equation

Existence of a solution to the heat equation

Find $u : \mathbb{R}_+ \times \mathbb{R}^d \to \mathbb{R}$ solution to

$$\begin{cases} \partial_t u - \Delta u = f \text{ in } \mathbb{R}_+ \times \mathbb{R}^d \\ u(0, \cdot) = u_0 \text{ in } \mathbb{R}^d \end{cases}$$

Two steps:

- The fundamental solution of the heat equation (also known as the heat kernel)
- A solution to the heat equation in $\mathbb{R}^d$ built by convolution against the fundamental solution

The heat kernel: formula

The heat kernel (in dimension $d \geq 1$) is defined by:

$$\Phi(t,x) = \begin{cases} \dfrac{1}{(4\pi t)^{d/2}} \exp\left(-\dfrac{|x|^2}{4t}\right) & \text{for } t > 0, x \in \mathbb{R}^d \\ 0 \text{ for } t \leq 0, x \in \mathbb{R}^d \end{cases}$$

$\rightarrow$ A Python code to see the heat kernel in action

Remark: The heat kernel can be derived by looking for a radially symmetric solution such that $u(\lambda^2 t, \lambda x) = c(t)u(t,x)$ (self-similar in space solution)

Remark: One may recognize the density of $\sqrt{2t}(G_1, \ldots, G_d)$ where $G_1, \ldots, G_d$ are i.i.d. Gaussian random variables with zero mean and variance one. There is a deep connection between the heat equation and probabilistic objects (Brownian motion)

The heat kernel: properties

- For any $t > 0$, $\displaystyle\int_{\mathbb{R}^d} \Phi(t,x)\,dx = 1$ (since $\int_{\mathbb{R}} \exp(-u^2)\,du = \sqrt{\pi}$)
- The function $\Phi : \mathbb{R} \times \mathbb{R}^d \setminus \{(0,0)\} \to \mathbb{R}$ is $\mathcal{C}^\infty$ and

$$\forall (t,x) \in \mathbb{R} \times \mathbb{R}^d \setminus \{(0,0)\}, \quad (\partial_t \Phi - \Delta \Phi)(t,x) = 0$$

- The function Φ is unbounded in every neighborhood of $(0,0)$ but it is locally integrable:

$$\int_{-1}^{1} \int_{B(0,1)} \Phi(t,x)\,dx\,dt < \infty$$

Indeed (using the change of variable $y = x/\sqrt{4t}$)

$$\begin{aligned}
\int_{-1}^{1} \int_{B(0,1)} \Phi(t,x)\,dx\,dt &= \int_0^1 \frac{1}{(4\pi t)^{d/2}} \int_{B(0,1)} \exp\left(-\frac{|x|^2}{4t}\right) dx\,dt \\
&= \frac{1}{\pi^{d/2}} \int_0^1 \int_{B(0,1/\sqrt{4t})} \exp\left(-|y|^2\right) dy\,dt \\
&\leq \frac{1}{\pi^{d/2}} \int_0^1 \int_{\mathbb{R}^d} \exp\left(-|y|^2\right) dy\,dt = 1 < \infty
\end{aligned}$$

The heat kernel: properties

Proposition

For any $\varphi \in \mathcal{C}_c^{1,2}(\mathbb{R} \times \mathbb{R}^d)$

$$\int_{\mathbb{R}} \int_{\mathbb{R}^d} \Phi(t,x) \left(-\partial_t \varphi - \Delta \varphi\right)(t,x)\, dx\, dt = \varphi(0,0)$$

Proof:

- $\int_{\mathbb{R}} \int_{\mathbb{R}^d} \Phi \left(-\partial_t \varphi - \Delta \varphi\right) = \lim_{\varepsilon \to 0} \int_{\varepsilon}^{\infty} \int_{\mathbb{R}^d} \Phi \left(-\partial_t \varphi - \Delta \varphi\right)$
- By integration by parts:
  $\int_{\varepsilon}^{\infty} \int_{\mathbb{R}^d} \Phi \left(-\partial_t \varphi - \Delta \varphi\right) = \int_{\varepsilon}^{\infty} \int_{\mathbb{R}^d} \left(\partial_t \Phi - \Delta \Phi\right) \varphi +$
  $\int_{\mathbb{R}^d} \Phi(\varepsilon, x) \varphi(\varepsilon, x)\, dx = \int_{\mathbb{R}^d} \frac{1}{(4\pi\varepsilon)^{d/2}} \exp\left(-\frac{|x|^2}{4\varepsilon}\right) \varphi(\varepsilon, x)\, dx$
- Change of variable ($y = x/\sqrt{4\varepsilon}$) and dominated convergence:
  $\int_{\mathbb{R}^d} \frac{1}{(4\pi\varepsilon)^{d/2}} \exp\left(-\frac{|x|^2}{4\varepsilon}\right) \varphi(\varepsilon, x)\, dx =$
  $\int_{\mathbb{R}^d} \frac{1}{\pi^{d/2}} \exp\left(-|y|^2\right) \varphi(\varepsilon, \sqrt{4\varepsilon} y)\, dy \xrightarrow[\varepsilon \to 0]{} \varphi(0,0)$

The same result written in distributional sense

In modern notation, one writes:

$$\partial_t \Phi - \Delta \Phi = \delta_{(0,0)}$$

in the sense of distributions: for any smooth and compactly supported test function φ, $\langle \partial_t \Phi - \Delta \Phi, \varphi \rangle = \langle \delta_{(0,0)}, \varphi \rangle$ which means:

$$\int_{\mathbb{R}} \int_{\mathbb{R}^d} \Phi \left(-\partial_t \varphi - \Delta \varphi \right) = \varphi(0,0)$$

Another formulation: $\lim_{t \to 0^+} \int_{\mathbb{R}^d} \Phi(t,x)\varphi(x)\, dx = \varphi(0)$ and thus

$$\begin{cases} \partial_t \Phi - \Delta \Phi = 0 \text{ for } t > 0, x \in \mathbb{R}^d \\ \quad \text{" } \Phi(0,\cdot) = \delta_0 \text{ on } \mathbb{R}^d \text{ "} \end{cases}$$

$\rightarrow$ more on distributions in the forthcoming lectures "Analyse 2"

A solution to the heat equation in $\mathbb{R}^d$ for $f = 0$

Theorem

Let u_0 be a bounded continuous function in $\mathbb{R}^d$. The function

$$u(t,x) = \int_{\mathbb{R}^d} \Phi(t, x-y) u_0(y)\, dy$$

solves the heat equation with $f = 0$ in the following sense:

- $u \in \mathcal{C}^\infty(\mathbb{R}_+^* \times \mathbb{R}^d)$
- *for all $t > 0$ and $x \in \mathbb{R}^d$, $\partial_t u(t,x) - \Delta u(t,x) = 0$*
- *for all $x \in \mathbb{R}^d$, $\lim_{t \to 0^+} u(t,x) = u_0(x)$*

Proof: (i) u is well defined on $\mathbb{R}_+^* \times \mathbb{R}^d$ and by dominated convergence, for $t > 0$ and $x \in \mathbb{R}^d$,

$$(\partial_t u - \Delta u)(t,x) = \int_{\mathbb{R}^d} (\partial_t - \Delta_x)\Phi(t, x-y) u_0(y)\, dy = 0$$

(ii) Change of variable ($z = (x-y)/\sqrt{4t}$) and dominated convergence:

$$\int_{\mathbb{R}^d} \frac{1}{(4\pi t)^{d/2}} e^{-\frac{|x-y|^2}{4t}} u_0(y)\, dy = \int_{\mathbb{R}^d} \frac{1}{\pi^{d/2}} e^{-|z|^2} u_0(x - \sqrt{4t}z)\, dz \xrightarrow[t \to 0]{} u_0(x)$$

A solution to the heat equation in $\mathbb{R}^d$ for $f \neq 0$

Remember the Duhamel principle for ordinary differential equations. The solution to $\dot{x}(t) = Ax(t)$ is $x(t) = \exp(At)x_0$. The solution to $\dot{x}(t) = Ax(t) + b(t)$ is $x(t) = \exp(At)x_0 + \int_0^t \exp(A(t-s))b(s)\,ds$

Theorem

Let u_0 be a bounded continuous function on $\mathbb{R}^d$ and $f \in \mathcal{C}^{1,2}(\mathbb{R}_+ \times \mathbb{R}^d)$ with compact support in $\mathbb{R}_+ \times \mathbb{R}^d$. Then

$$u(t,x) = \int_{\mathbb{R}^d} \Phi(t, x-y)u_0(y)\,dy + \int_0^t \int_{\mathbb{R}^d} \Phi(t-s, x-y)f(s,y)\,dy\,ds$$

solves the heat equation in the following sense:

- $u \in \mathcal{C}^{1,2}(\mathbb{R}_+^* \times \mathbb{R}^d)$
- *for all $t > 0$ and $x \in \mathbb{R}^d$, $\partial_t u(t,x) - \Delta u(t,x) = f(t,x)$*
- *for all $x \in \mathbb{R}^d$, $\lim_{t \to 0^+} u(t,x) = u_0(x)$*

Towards qualitative properties

We identified one solution

$$u(t,x) = \int_{\mathbb{R}^d} \Phi(t, x-y) u_0(y)\, dy + \int_0^t \int_{\mathbb{R}^d} \Phi(t-s, x-y) f(s,y)\, dy\, ds \tag{2}$$

to

$$\begin{cases} \partial_t u - \Delta u = f \\ u(0,\cdot) = u_0 \end{cases}$$

Questions:

- Uniqueness: Is the convolution (2) the unique solution? This is related to a maximum principle
- Weak maximum principle: If $u_0 \leq 0$ and $f \leq 0$, then (2) is non positive. Is this true for any solution of the heat equation?
- Regularization: If $f = 0$ and u_0 is bounded continuous, $u(t,\cdot)$ defined by (2) is $\mathcal{C}^\infty(\mathbb{R}^d)$ for any $t > 0$. Is this true for any solution?

Weak maximum principle on a bounded subset

Theorem

Let $T > 0$ and $\Omega \subset \mathbb{R}^d$ be an open bounded subset of $\mathbb{R}^d$, and let $u \in C^{1,2}((0,T] \times \Omega) \cap C([0,T] \times \overline{\Omega})$ satisfy

$$\partial_t u - \Delta u \leq 0 \quad \text{in } (0,T] \times \Omega$$

Let

$$\Gamma_T = \{0\} \times \Omega \cup [0,T] \times \partial\Omega$$

denote the so-called parabolic boundary of $(0,T] \times \Omega$. Then

$$\max_{[0,T] \times \overline{\Omega}} u = \max_{\Gamma_T} u$$

The weak maximum principle is equivalent to (set $\tilde{u} = u - \max_{\Gamma_T} u$)

$$u \leq 0 \text{ on } \Gamma_T \Rightarrow u \leq 0 \text{ on } [0,T] \times \overline{\Omega}$$

This has a very natural physical interpretation!

Weak maximum principle on a bounded subset: proof

Proof:

- If $\partial_t u - \Delta u < 0$ in $(0, T] \times \Omega$ and there exists $(t_0, x_0) \in [0, T] \times \overline{\Omega} \setminus \Gamma_T$ s. t. $u(t_0, x_0) = \max_{[0,T]\times\overline{\Omega}} u$, then $\partial_t u(t_0, x_0) \geq 0$ and $\Delta u(t_0, x_0) \leq 0 \rightarrow$ contradiction.
- Assume now $\partial_t u - \Delta u \leq 0$ and consider $v_\varepsilon(t, x) = u(t, x) - \varepsilon t$. One has $\partial_t v_\varepsilon - \Delta v_\varepsilon < 0$, thus $\max_{[0,T]\times\overline{\Omega}} v_\varepsilon = v_\varepsilon(t_\varepsilon, x_\varepsilon)$ for some $(t_\varepsilon, x_\varepsilon) \in \Gamma_T$.
  Since Γ_T is compact, $\exists(\varepsilon_n)_{n\geq 0}$ s.t. $\varepsilon_n \xrightarrow[n\to\infty]{} 0$ and $(t_{\varepsilon_n}, x_{\varepsilon_n}) \xrightarrow[n\to\infty]{} (t^*, x^*) \in \Gamma_T$. Send $n \to \infty$ in the inequality: for any $(t, x) \in [0, T] \times \overline{\Omega}$,

$$u(t, x) - \varepsilon_n t \leq v_{\varepsilon_n}(t_{\varepsilon_n}, x_{\varepsilon_n}) \leq u(t_{\varepsilon_n}, x_{\varepsilon_n})$$

  to get $\max_{[0,T]\times\overline{\Omega}} u = u(t^*, x^*)$

Weak maximum principle on $\mathbb{R}^d$

On $(0, T] \times \mathbb{R}^d$, the parabolic boundary is simply $\{0\} \times \mathbb{R}^d$. Do we have $\sup_{[0,T]\times\mathbb{R}^d} u = \sup_{\mathbb{R}^d} u_0$ without any "boundary condition"?
$\rightarrow$ No... "Heat can come from infinity!"

Theorem

Let $T > 0$ and let $u \in C^{1,2}((0, T] \times \mathbb{R}^d) \cap C([0, T] \times \mathbb{R}^d)$ satisfy

$$\begin{cases} \partial_t u - \Delta u \leq 0 \text{ in } (0, T] \times \mathbb{R}^d \\ u(0, x) = u_0(x) \text{ for } x \in \mathbb{R}^d. \end{cases}$$

Let us moreover assume that $\exists a, A > 0$ s.t., $\forall (t, x) \in [0, T] \times \mathbb{R}^d$,

$$u(t, x) \leq A \exp(a|x|^2)$$

Then

$$\sup_{[0,T]\times\mathbb{R}^d} u = \sup_{\mathbb{R}^d} u_0$$

Corollary: a uniqueness result

Corollary

Let u_0 be bounded and continuous and $f \in \mathcal{C}_c^{1,2}(\mathbb{R}_+ \times \mathbb{R}^d)$. Let u be a solution to the heat equation

$$\begin{cases} \partial_t u - \Delta u = f \\ u(0,\cdot) = u_0 \end{cases}$$

satisfying the growth estimate:

$$\forall (t,x) \in \mathbb{R}_+ \times \mathbb{R}^d,\ |u(t,x)| \leq A \exp(a|x|^2)$$

for some positive constants a and A. Then

$$u(t,x) = \int_{\mathbb{R}^d} \Phi(t, x-y) u_0(y)\, dy + \int_0^t \int_{\mathbb{R}^d} \Phi(t-s, x-y) f(s,y)\, dy\, ds$$

Proof: (i) The solution built by convolution satisfies the growth estimate

(ii) Uniqueness via the weak maximum principle

Regularity of solutions to the homogeneous heat equation

Theorem

Let $\Omega \subset \mathbb{R}^d$ be an open set. Let us assume that $u \in \mathcal{C}^{1,2}((0,T) \times \Omega)$ satisfies the homogeneous heat equation:

$$\partial_t u - \Delta u = 0 \text{ in } (0,T) \times \Omega.$$

Then $u \in \mathcal{C}^\infty((0,T) \times \Omega)$.

- Notice that u can be non-smooth (not $\mathcal{C}^\infty$) on $\partial[(0,T) \times \Omega]$!
- Basic idea: even if u_0 is only bounded continuous, $\int_{\mathbb{R}^d} \Phi(t, x-y) u_0(y)\, dy$ is $C^\infty(\mathbb{R}_+^* \times \mathbb{R}^d)$. The argument consists in localizing the argument using a cutoff function

Two fundamental properties of the heat equation

Take-home messages:

- Instantaneous smoothing property: if u_0 is only bounded continuous, $\int_{\mathbb{R}^d} \Phi(t, x-y)u_0(y)\,dy$ is $C^\infty(\mathbb{R}_+^* \times \mathbb{R}^d)$. The solution is immediately smooth
- Transport of energy at infinite speed: Let $u_0 \geq 0$ be continuous, bounded, non-zero, and with compact support. Then the solution to the heat equation (for $f = 0$)

$$u(t,x) = \int_{\mathbb{R}^d} \Phi(t, x-y)u_0(y)\,dy$$

  is (strictly) positive for all $t > 0$ and $x \in \mathbb{R}^d$. The heat equation propagates heat (energy) at infinite speed

Remark: We did not discuss the existence of solutions to the heat equation on a bounded domain. The approach using kernels is cumbersome $\rightarrow$ see "Equations aux dérivées partielles: approches variationnelles" for better techniques!

Discretization using finite differences

Let us consider the heat equation in dimension 1, on $\Omega = (0,1)$, with Dirichlet boundary conditions

$$\begin{cases} \partial_t u - \partial_{x,x} u = f \text{ on } (0,T) \times \Omega \\ u(t,0) = a \text{ and } u(t,1) = b \text{ for } t \in (0,T) \\ u(0,x) = u_0(x) \text{ for } x \in \Omega \end{cases}$$

We assume there exists a unique solution $u \in \mathcal{C}^{1,2}((0,T) \times \Omega) \cap \mathcal{C}^0([0,T] \times \overline{\Omega})$.

Remark: As for the Poisson equation, we stick to dimension $d = 1$ to keep things simple: everything can be generalized to higher dimensions (but keep in mind the curse of dimensionality)

Discretization in space by the finite difference method

Step 1: Discretization in space. Introduce a discretization of $(0,1)$ into I intervals: $(x_i)_{0\leq i\leq I}$ where $x_i = ih$ and $h = \frac{1}{I}$

We already know $u(t,x_0) = a$ and $u(t,x_I) = b$. Write the heat equation at the interior points: for $t > 0$,

$$\partial_t u(t,x_i) - \partial_{x,x} u(t,x_i) = f(t,x_i) \text{ for } i \in \{1,\dots,I-1\}$$

Approximate the differential operator in space using finite differences: if $u(t,\cdot) \in \mathcal{C}^4(\overline{\Omega})$,

$$\partial_{x,x} u(t,x_i) = D_h^2 u(t,x_i) + O(h^2)$$

where $D_h^2 u(t,x_i) = \dfrac{u(t,x_{i+1}) - 2u(t,x_i) + u(t,x_{i-1})}{h^2}$

The space discretized problem

One gets the space discretized problem: find $U : \mathbb{R}_+ \to \mathbb{R}^{I-1}$ s.t.

$$\begin{cases} \dot{U}(t) = -AU(t) + C + F(t) \text{ for } t \in (0, T) \\ U(0) = U^0 \end{cases}$$

with $A \in \mathbb{R}^{(I-1)\times(I-1)}$, $C \in \mathbb{R}^{I-1}$, $F(t) \in \mathbb{R}^{I-1}$ and $U^0 \in \mathbb{R}^{I-1}$:

$$A = \frac{1}{h^2} \begin{bmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{bmatrix}$$

$$C = \frac{1}{h^2} \begin{pmatrix} a \\ 0 \\ \vdots \\ 0 \\ b \end{pmatrix} \quad F(t) = \begin{pmatrix} f(t, x_1) \\ f(t, x_2) \\ \vdots \\ f(t, x_{I-2}) \\ f(t, x_{I-1}) \end{pmatrix} \text{ and } U^0 = \begin{pmatrix} u_0(x_1) \\ u_0(x_2) \\ \vdots \\ u_0(x_{I-2}) \\ u_0(x_{I-1}) \end{pmatrix}$$

Time discretization

The ordinary differential equation on U is then discretized using standard schemes: $\Delta t = T/N$ and $n \in \{0, \dots, N-1\}$

- Explicit Euler scheme: (order 1 approximation)

$$\frac{U(t_{n+1}) - U(t_n)}{\Delta t} \approx \dot{U}(t_n) = -AU(t_n) + C + F(t_n)$$

which yields

$$U^{n+1} = (\mathrm{Id} - \Delta t A)U^n + \Delta t C + \Delta t F(t_n)$$

- Implicit Euler scheme: (order 1 approximation)

$$\frac{U(t_{n+1}) - U(t_n)}{\Delta t} \approx \dot{U}(t_{n+1}) = -AU(t_{n+1}) + C + F(t_{n+1})$$

which yields

$$(\mathrm{Id} + \Delta t A)\, U^{n+1} = U^n + \Delta t C + \Delta t F(t_{n+1})$$

Time discretization

- Crank-Nicolson scheme: (order 2 approximation)

$$\frac{U(t_{n+1}) - U(t_n)}{\Delta t} \approx \dot{U}(t_{n+1/2}) = -A\frac{U(t_n) + U(t_{n+1})}{2} + C + \frac{F(t_n) + F(t_{n+1})}{2}$$

which yields an average of the two previous schemes

$$\left(\mathrm{Id} + \frac{\Delta t}{2}A\right) U^{n+1} = \left(\mathrm{Id} - \frac{\Delta t}{2}A\right) U^n + \Delta t C + \frac{\Delta t}{2}\left(F(t_n) + F(t_{n+1})\right)$$

Remark: Recall that A is symmetric positive definite so that all these schemes are well defined (the matrix $(\mathrm{Id} + \Delta t A)$ is invertible for any $\Delta t > 0$)

Error analysis

Let us introduce the error vectors: for all $n \in \{0, \dots, N\}$

$$E^n = \pi(u(t_n, \cdot)) - U^n \in \mathbb{R}^{I-1}$$

where, we recall, $\forall i \in \{1, \dots, I-1\}$,

$$[\pi(u(t_n, \cdot))]_i = u(t_n, x_i)$$

We would like to study the asymptotics of $\sup_{0 \le n \le N} \|E^n\|$ in the limit $(\Delta t, h) \to (0, 0)$, where $\|\cdot\|$ is either

- The l^∞-norm: $\|E^n\|_\infty = \sup_{1 \le i \le I-1} |E_i^n|$
- The l^2-norm: $\|E^n\|_2 = \sqrt{\frac{1}{I-1} \sum_{i=1}^{I-1} |E_i^n|^2}$

The convergence analysis relies on two steps:

- A consistency analysis: what are the errors when one plugs the solution u in the numerical scheme?
- A stability analysis: how do the consistency errors propagate to the global errors $\|E^n\|$?

Consistency error: the explicit Euler scheme

Let us consider e.g. the explicit Euler scheme:

$$\frac{U^{n+1} - U^n}{\Delta t} = -AU^n + C + F(t_n)$$

The consistency error vectors ε^n are defined by: for $n \in \{0, \dots, N-1\}$,

$$\frac{\pi(u(t_{n+1}, \cdot)) - \pi(u(t_n, \cdot))}{\Delta t} = -A\pi(u(t_n, \cdot)) + C + F(t_n) + \varepsilon^n$$

If u is sufficiently regular ($u \in \mathcal{C}^{2,4}([0, T] \times \overline{\Omega})$), one has:

$$\sup_{0 \le n \le N-1} \|\varepsilon^n\|_\infty \le C(\Delta t + h^2)$$

Consistency error: implicit schemes

Likewise, for the Implicit Euler scheme, ε^n is defined by: for $n \in \{0, \dots, N-1\}$,

$$\frac{\pi(u(t_{n+1}, \cdot)) - \pi(u(t_n, \cdot))}{\Delta t} = -A\pi(u(t_{n+1}, \cdot)) + C + F(t_{n+1}) + \varepsilon^n$$

and for $u \in \mathcal{C}^{2,4}([0, T] \times \overline{\Omega})$,

$$\sup_{0 \le n \le N-1} \|\varepsilon^n\|_\infty \le C(\Delta t + h^2)$$

For the Crank-Nicolson scheme, ε^n is defined by: for $n \in \{0, \dots, N-1\}$,

$$\frac{\pi(u(t_{n+1}, \cdot)) - \pi(u(t_n, \cdot))}{\Delta t} = -A\frac{\pi(u(t_n, \cdot)) + \pi(u(t_{n+1}, \cdot))}{2} + C + \frac{F(t_n) + F(t_{n+1})}{2} + \varepsilon^n$$

and for $u \in \mathcal{C}^{4,4}([0, T] \times \overline{\Omega})$,

$$\sup_{0 \le n \le N-1} \|\varepsilon^n\|_\infty \le C(\Delta t^2 + h^2)$$

Stability analysis: the explicit Euler scheme

Again, let us consider the explicit Euler scheme:

$$\frac{U^{n+1} - U^n}{\Delta t} = -AU^n + C + F(t_n)$$

$$\frac{\pi(u(t_{n+1}, \cdot)) - \pi(u(t_n, \cdot))}{\Delta t} = -A\pi(u(t_n, \cdot)) + C + F(t_n) + \varepsilon^n$$

and thus

$$\frac{E^{n+1} - E^n}{\Delta t} = -AE^n + \epsilon^n$$

i.e.

$$E^{n+1} = (\mathrm{Id} - \Delta t A)E^n + \Delta t \varepsilon^n$$

Therefore (since $E^0 = 0$), for all $n \in \{1, \dots, N\}$,

$$E^n = \Delta t \sum_{k=0}^{n-1} (\mathrm{Id} - \Delta t A)^{n-1-k} \varepsilon^k$$

Stability analysis: the explicit Euler scheme

Recall the induced matrix norm: for $M \in \mathbb{R}^{n\times n}$,
$\|M\| = \sup_{x\in\mathbb{R}^n, \|x\|\leq 1} \|Mx\|$

From $E^n = \Delta t \sum_{k=0}^{n-1} (\mathrm{Id} - \Delta t A)^{n-1-k} \varepsilon^k$ and
$\sup_{0\leq n\leq N-1} \|\varepsilon^n\|_\infty \leq C(\Delta t + h^2)$, one gets:

$$\|E^n\| \leq \Delta t \sum_{k=0}^{n-1} \|\mathrm{Id} - \Delta t A\|^{n-1-k} \|\varepsilon^k\|$$

$$\leq C\Delta t \left(\sum_{k=0}^{n-1} \|\mathrm{Id} - \Delta t A\|^{n-1-k} \right) (\Delta t + h^2)$$

$$= C\Delta t \left(\sum_{k=0}^{n-1} \|\mathrm{Id} - \Delta t A\|^{k} \right) (\Delta t + h^2)$$

The numerical scheme is stable if $\|\mathrm{Id} - \Delta t A\| \leq 1$ in which case

$$\forall n \in \{0, \ldots, N\}, \quad \|E^n\| \leq CT(\Delta t + h^2)$$

Lax principle: consistency + stability $\Rightarrow$ convergence

Stability analysis

Likewise, for the implicit Euler scheme, one has

$$E^n = \Delta t \sum_{k=0}^{n-1} [(\mathrm{Id} + \Delta t A)^{-1}]^{n-k} \varepsilon^k$$

and thus, the numerical scheme is stable if

$$\|(\mathrm{Id} + \Delta t A)^{-1}\| \leq 1$$

For the Crank-Nicolson scheme, one has

$$E^n = \Delta t \sum_{k=0}^{n-1} \left[\left(\mathrm{Id} + \frac{\Delta t}{2} A \right)^{-1} \left(\mathrm{Id} - \frac{\Delta t}{2} A \right) \right]^{n-1-k} \left(\mathrm{Id} + \frac{\Delta t}{2} A \right)^{-1} \varepsilon^k$$

and thus, the numerical scheme is stable if

$$\left\| \left(\mathrm{Id} + \frac{\Delta t}{2} A \right)^{-1} \left(\mathrm{Id} - \frac{\Delta t}{2} A \right) \right\| \leq 1 \text{ and } \left\| \left(\mathrm{Id} + \frac{\Delta t}{2} A \right)^{-1} \right\| \leq 1$$

Stability in l^2-norm: reminder

Remember that for a symmetric positive matrix S, $\|S\|_2 = \max\{|\lambda|, \lambda \text{ eigenvalue of } S\}$. Let us also recall the Lemma

Lemma

Let us consider $\tilde{A} \in \mathbb{R}^{(I-1)\times(I-1)}$*:*

$$\tilde{A} = \begin{bmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{bmatrix}$$

The eigenvalues of $\tilde{A}$ *are* $\lambda_i = 4\sin^2\left(\frac{i\pi}{2I}\right)$ $(1 \leq i \leq I-1)$ *with eigenvectors* $v_i \in \mathbb{R}^{I-1}$ *defined by* $\forall j \in \{1, \ldots, I-1\}, [v_i]_j = \sin\left(\frac{ij\pi}{I}\right)$

Stability in l^2-norm: explicit Euler

Thus, for the explicit Euler scheme, the eigenvalues of $\mathrm{Id} - \Delta t A$ are

$$\left(1 - \Delta t \frac{\lambda_i}{h^2}\right)_{1 \leq i \leq I-1}$$

and thus, $\rho(\mathrm{Id} - \Delta t A) \leq 1$ is equivalent to

$$\Delta t \leq \frac{2h^2}{\lambda_{I-1}}$$

Since $\lambda_{I-1} \leq 4$ and $\lim_{I \to \infty} \lambda_{I-1} = 4$, it is thus natural to enforce the stronger condition:

$$\Delta t \leq \frac{h^2}{2} \tag{3}$$

Proposition

The explicit Euler scheme is stable in l^2-norm under the Courant-Friedrichs-Lewy (CFL) condition (3)

Stability in l^2-norm: implicit schemes

The explicit Euler scheme is thus only conditionally stable in l^2-norm

In contrast, for the implicit Euler scheme, the eigenvalues of $\mathrm{Id} + \Delta t A$ are $\left(1 + \Delta t \frac{\lambda_i}{h^2}\right)_{1 \leq i \leq I-1}$, and thus

$$\rho((\mathrm{Id} + \Delta t A)^{-1}) = \left(1 + \Delta t \frac{\lambda_1}{h^2}\right)^{-1} < 1$$

for any Δt and h. The implicit Euler scheme is thus unconditionally stable in l^2-norm

Likewise, the Crank-Nicolson scheme is unconditionally stable in l^2-norm

Conclusion

Theorem

- *The explicit Euler scheme is convergent under the CFL condition* $\Delta t \leq \frac{h^2}{2}$*, with order* 1 *in time and* 2 *in space, in the* l^2*- and* l^∞*-norms.*
- *The implicit Euler scheme is convergent with order* 1 *in time and* 2 *in space, in the* l^2*- and* l^∞*-norms.*
- *The Crank-Nicolson scheme is convergent with order* 2 *in time and* 2 *in space, unconditionally in the* l^2*-norm, and under a CFL condition in the* l^∞*-norm.*

Take-home message: implicit schemes are more stable, but at the expense of more computations at each time-step (solving a linear system). For parabolic problems such as the heat equation, the error versus computational cost balance is in general in favor of implicit schemes